DOTNET INTERVIEW QUESTIONS

C# Fundamentals

1. What is the difference between a struct and a class in C#?
2. Explain how async and await work under the hood and why blocking on .Result can deadlock an ASP.NET application that uses a synchronization context?
3. What are the SOLID principles?
4. Describe dependency injection lifetimes in ASP.NET Core.

SQL Server

1. What is the difference between clustered and non-clustered indexes?
2. How do you find and fix a slow query in SQL Server?
3. Explain isolation levels and the problems each one prevents.

System Design

1. Design a URL shortener that handles 1000 writes per second.
2. Design a rate limiter for a public API.